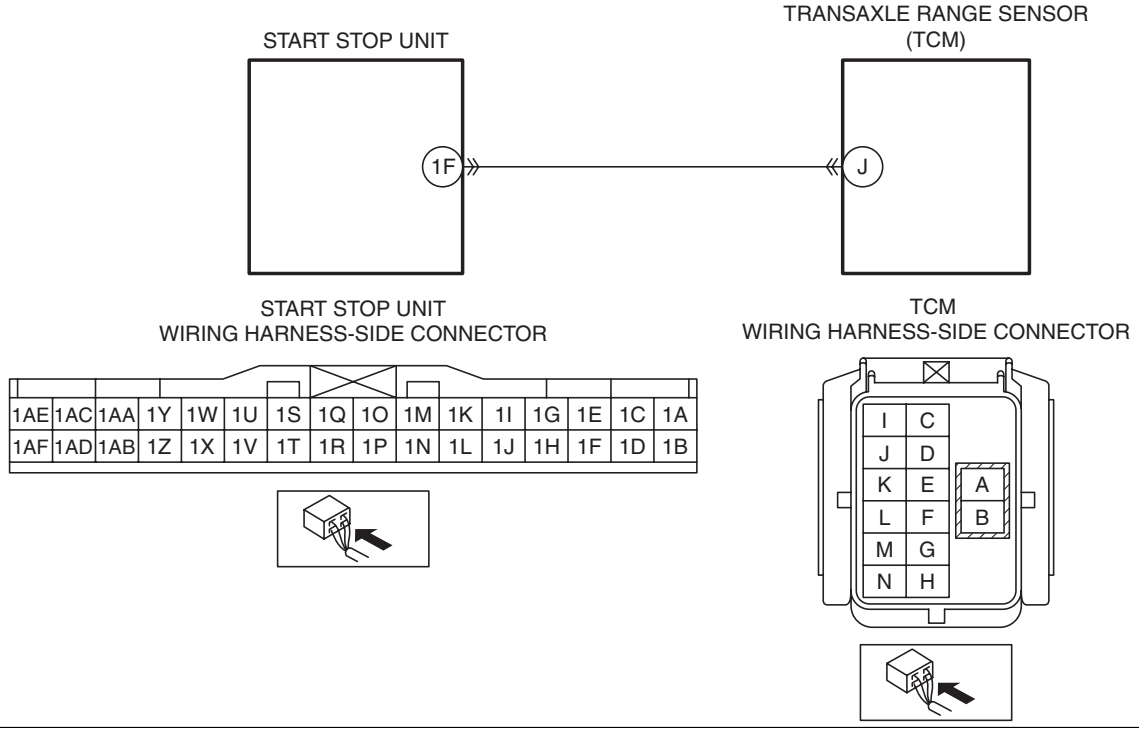
DTC P0850:23 [START STOP UNIT]

id0902p6023200

Description	Transaxle range sensor circuit malfunction
Detection condition	• The start stop unit detected the following condition **5 times** continuously with the ignition switched ON (engine off or on). — A transaxle range sensor signal is in the P or N position for a continuous **1 s or more** even though the TCM CAN signal is in a position other than the P and N positions.
Fail-safe	• Perform control of the internal signal for control of the transaxle range sensor as position other than P, N positions.
Possible cause	• DTCs are stored in the TCM. • Start stop unit connector or terminal malfunction • TCM connector or terminal malfunction • Short to power supply in wiring harness between start stop unit terminal 1F and TCM terminal J • Transaxle range sensor malfunction • Start stop unit malfunction



Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY TCM DTCs** • Retrieve the TCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (CW6A-EL)].) (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (EW6A-EL)].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See ON-BOARD DIAGNOSTIC SYSTEM DTC TABLE [TCM (CW6A-EL)].) (See ON-BOARD DIAGNOSTIC SYSTEM DTC TABLE [TCM (EW6A-EL)].)
		No	Go to the next step.
2	**INSPECT START STOP UNIT CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the start stop unit connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 6.

Step	Inspection		Action
3	**INSPECT TCM CONNECTOR CONDITION** • Disconnect the TCM connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 6.
4	**INSPECT TRANSAXLE RANGE SENSOR CIRCUIT FOR SHORT TO POWER SUPPLY** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Operate the selector lever to P or N position. • Measure the voltage at the start stop unit terminal 1F (wiring harness-side). • Is the voltage **1.0 V or less**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between start stop unit terminal 1F and TCM terminal J. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 6.
5	**VERIFY IF MALFUNCTIONING LOCATION IS TRANSAXLE RANGE SENSOR DEPENDING ON REPEATABILITY** • Clear the DTC for the start stop unit using the M-MDS. (See CLEARING DTC [START STOP UNIT].) • Switch the ignition ON (engine off). • Perform the following work **5 times or more**. — Operate the selector lever to P or N position and wait for **1 s or more**. — Operate the selector lever to other than P and N position and wait for **1 s or more**. — Operate the selector lever to P or N position. • Retrieve the start stop unit DTCs using the M-MDS. (See DTC INSPECTION [START STOP UNIT].) • Is the same DTC displayed?	Yes	Replace the TCM, then go to the next step. (See CONTROL VALVE BODY REMOVAL/INSTALLATION [CW6A-EL].) (See CONTROL VALVE BODY REMOVAL/INSTALLATION [EW6A-EL].)
		No	Go to Step 7.
6	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the start stop unit using the M-MDS. (See CLEARING DTC [START STOP UNIT].) • Switch the ignition ON (engine off). • Perform the following work **5 times or more**. — Operate the selector lever to P or N position and wait for **1 s or more**. — Operate the selector lever to other than P and N position and wait for **1 s or more**. — Operate the selector lever to P or N position. • Retrieve the start stop unit DTCs using the M-MDS. (See DTC INSPECTION [START STOP UNIT].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the start stop unit. (See START STOP UNIT REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.

Step	Inspection		Action
7	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [START STOP UNIT].)
		No	DTC troubleshooting completed.